

Completeness of Higher Baire-Class Function Spaces: An Abstract Quasi-Completion Theorem and the Metrizable Program

Candidate partial result for arXiv:2601.00733

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Status: candidate partial result, likely valid. The strong completeness/reflexivity half is settled for arbitrary Tychonoff spaces, both halves are settled for metrizable spaces, and the full Baire union $B(X)$ is settled. The fixed-rank local-completeness problem for arbitrary nonmetrizable spaces remains open.

1 Source problem

Gabrielyan, Osipov, and Reznichenko characterize completeness and reflexivity-type properties of the pointwise space $B_1(X)$ and then ask for the corresponding program for every countable $\alpha > 1$ and for $B(X) = \bigcup_{\alpha < \omega_1} B_\alpha(X)$ [1, Problem 3.10, PDF p. 11]. The exact problem is reproduced below.

Example 3.9 ([30, Corollary 4.6]). *There is a separable pseudocompact space X such that $B_1(X)$ is a Baire space which is not Choquet.*

We finish this section with the following natural open problem.

Problem 3.10. *Let $\alpha > 1$ be a countable ordinal. Characterize spaces X for which $B_\alpha(X)$ has one of the properties from Theorems 3.3 and 3.4. Analogously for the space $B(X)$.*

4. Baire-one functions on separable metrizable spaces

In this section, we consider the important case of separable metrizable spaces. It turns out that, in the realm of separable metrizable spaces, the results obtained in the previous section lead us to some classical classes of separable metric spaces which were considered, for example, in [57, 111].

Figure 1: Problem 3.10 in arXiv:2601.00733v2, PDF page 11.

All function spaces below carry the topology of pointwise convergence, inherited from $\mathbb{R}^X$. We use the source's definitions of complete, quasi-complete, (semi-)reflexive, (semi-)Montel, sequentially complete, and locally complete. In particular, local completeness means that the closed absolutely convex hull of every null sequence is compact.

For a nonzero countable ordinal α , $B_\alpha(X)$ is the usual Baire class: $B_0(X) = C(X)$ and $B_\alpha(X)$ consists of pointwise limits of sequences from $\bigcup_{\xi < \alpha} B_\xi(X)$. For a metrizable X , write $\Sigma_\xi^0(X)$ and $\Pi_\xi^0(X)$ for the additive and multiplicative Borel classes.

2 Proof intuition

There are two different mechanisms.

First, quasi-completeness is so strong in a pointwise function space that it has nothing specifically to do with Baire rank. Continuous functions can interpolate arbitrary zero-one data on each finite subset of a Tychonoff space. The resulting net is pointwise Cauchy and stays in the bounded unit box. If the ambient pointwise subspace is quasi-complete, the net must converge inside it, so every characteristic function belongs to the space. Bounded finite-range approximations then put every real function in the space. Thus any quasi-complete pointwise vector space containing $C(X)$ is already the whole product $\mathbb{R}^X$.

Second, on a metrizable domain the Lebesgue–Hausdorff theorem identifies Baire rank α with Borel rank $\Sigma_{\alpha+1}^0$. If A has additive rank $\alpha + 1$, it can be partitioned into disjoint ambiguous-rank- $\alpha + 1$ pieces A_n . The functions $2^{n+1}\mathbf{1}_{A_n}$ form a null sequence. Local completeness makes its closed disk compact, while the disk contains partial sums converging to $\mathbf{1}_A$. Hence $\mathbf{1}_A$ already has Baire rank α , forcing the additive and multiplicative Borel classes to coincide. That equality collapses the next Baire class and gives sequential completeness.

3 The abstract quasi-completion theorem

Theorem 1 (Pointwise quasi-completion). *Let X be a Tychonoff space and let E be a vector subspace of $\mathbb{R}^X$, endowed with the pointwise topology, such that $C(X) \subseteq E$. Then*

$$E \text{ is quasi-complete} \iff E = \mathbb{R}^X.$$

Proof. Only the forward implication requires proof. Fix an arbitrary set $A \subseteq X$. Direct the finite subsets $F \subseteq X$ by inclusion. For $a \in F \cap A$, complete regularity gives a function $v_{a,F} \in C(X, [0, 1])$ such that

$$v_{a,F}(a) = 1, \quad v_{a,F}(F \setminus \{a\}) = \{0\}.$$

Set $u_F = 0$ if $F \cap A = \emptyset$, and otherwise set $u_F = \max_{a \in F \cap A} v_{a,F}$. Then

$$u_F \in C(X, [0, 1]), \quad u_F|_F = \mathbf{1}_A|_F.$$

The box

$$K = E \cap [0, 1]^X$$

is closed and bounded in E , hence complete by quasi-completeness. The net (u_F) lies in K and is pointwise Cauchy: once F contains a fixed $x \in X$, every later $u_G(x)$ equals $\mathbf{1}_A(x)$. Therefore (u_F) converges in K . Its pointwise limit is uniquely $\mathbf{1}_A$, so $\mathbf{1}_A \in E$. Since A was arbitrary, E contains every finite-range real function on X .

Now fix $f \in \mathbb{R}^X$. Define the finite-range function

$$s_n(x) = 2^{-n} \max\{-n2^n, \min\{n2^n, \lfloor 2^n f(x) \rfloor\}\}.$$

Then $s_n \in E$, $s_n(x) \rightarrow f(x)$ for every x , and $|s_n(x)| \leq |f(x)| + 1$. Thus (s_n) is a pointwise Cauchy sequence whose range is bounded in E . Its closure in E is a closed bounded set and is therefore complete. Hence (s_n) converges in E , and uniqueness of the coordinatewise limit gives $f \in E$. Thus $E = \mathbb{R}^X$.

The product $\mathbb{R}^X$ is complete, so the converse follows. $\square$

Corollary 2 (All strong properties). *Let X be Tychonoff and let E be either $B_\alpha(X)$, for a nonzero countable ordinal α , or $B(X)$. The following are equivalent:*

1. E is Montel;
2. E is semi-Montel;
3. E is reflexive;
4. E is semi-reflexive;
5. E is complete;
6. E is quasi-complete;
7. $E = \mathbb{R}^X$.

Proof. The source proves that $B_1(X)$ is barrelled [1, Theorem 1.1(i)]. A locally convex space containing a dense barrelled subspace, with the inherited topology on that subspace, is barrelled: if D is a closed barrel, D intersects the dense subspace in a neighborhood; shrinking that neighborhood in the ambient space and using density and closedness puts an ambient neighborhood inside D . Hence every E in the statement is barrelled.

Finite interpolation shows that $C(X)$, and therefore E , is pointwise dense in $\mathbb{R}^X$. Every continuous linear functional on E depends on finitely many coordinates, so the pointwise topology is precisely the weak topology of E . The standard theorem for barrelled locally convex spaces carrying their weak topology makes semi-Montel, reflexive, semi-reflexive, and quasi-complete equivalent; this is the same result used in the source [1, proof of Theorem 3.3] and comes from [2, Proposition 11.4.2]. Completeness implies quasi-completeness, and Theorem 1 says that quasi-completeness forces $E = \mathbb{R}^X$. Finally, $\mathbb{R}^X$ has every listed property. $\square$

4 Sequential completeness at an arbitrary rank

Theorem 3 (Successor-rank criterion). *For every Tychonoff space X and every nonzero countable ordinal α ,*

$$B_\alpha(X) \text{ is sequentially complete} \iff B_\alpha(X) = B_{\alpha+1}(X).$$

Proof. Because the Baire hierarchy is increasing, $B_{\alpha+1}(X)$ is exactly the set of pointwise limits of sequences from $B_\alpha(X)$. Every pointwise Cauchy sequence in $B_\alpha(X)$ has a coordinatewise limit in $\mathbb{R}^X$, and that limit belongs to $B_{\alpha+1}(X)$. Thus equality of the two classes is equivalent to retaining every Cauchy-sequence limit in $B_\alpha(X)$. $\square$

5 The complete metrizable program

We use the classical Lebesgue–Hausdorff theorem: if X is metrizable and $0 < \alpha < \omega_1$, then a real function f is of Baire class α if and only if $f^{-1}(U) \in \Sigma_{\alpha+1}^0(X)$ for every open $U \subseteq \mathbb{R}$; in particular,

$$\mathbf{1}_A \in B_\alpha(X) \iff A \in \Delta_{\alpha+1}^0(X) := \Sigma_{\alpha+1}^0(X) \cap \Pi_{\alpha+1}^0(X).$$

See, for example, the classical treatments in [4, 3].

Theorem 4 (Metrizability characterization). *Let X be metrizable and let $0 < \alpha < \omega_1$.*

(A) Strong properties. *Each of the seven equivalent conditions in Corollary 2 for $E = B_\alpha(X)$ is equivalent to*

$$\Sigma_{\alpha+1}^0(X) = \mathcal{P}(X),$$

that is, every subset of X has additive Borel rank at most $\alpha + 1$.

(B) Sequential and local completeness. *The following are equivalent:*

1. $B_\alpha(X)$ is sequentially complete;
2. $B_\alpha(X)$ is locally complete;
3. $B_\alpha(X) = B_{\alpha+1}(X)$;
4. $\Sigma_{\alpha+1}^0(X) = \Pi_{\alpha+1}^0(X)$.

Proof. For (A), Corollary 2 reduces everything to $B_\alpha(X) = \mathbb{R}^X$. If that equality holds, every characteristic function has Baire class α , so every subset lies in $\Delta_{\alpha+1}^0$. Conversely, if every subset lies in $\Sigma_{\alpha+1}^0$, then the inverse image of every open set under every $f : X \rightarrow \mathbb{R}$ lies in that class. The Lebesgue–Hausdorff theorem gives $f \in B_\alpha(X)$.

For (B), Theorem 3 gives (1) $\Leftrightarrow$ (3), and sequential completeness implies local completeness in every locally convex space.

Assume (2), and fix $A \in \Sigma_{\alpha+1}^0(X)$. Write $A = \bigcup_n C_n$ with $C_n \in \Pi_\alpha^0(X)$, and put $D_n = \bigcup_{k \leq n} C_k$. Finite unions preserve Π_α^0 . The sets

$$A_0 = D_0, \quad A_n = D_n \setminus D_{n-1} \quad (n \geq 1)$$

are pairwise disjoint, cover A , and belong to $\Delta_{\alpha+1}^0(X)$. Therefore $\mathbf{1}_{A_n} \in B_\alpha(X)$.

Set $g_n = 2^{n+1} \mathbf{1}_{A_n}$. Disjointness makes (g_n) a null sequence in the pointwise topology. By local completeness, its closed absolutely convex hull K is compact. For every N ,

$$F_N = \sum_{n \leq N} 2^{-n-1} g_n = \mathbf{1}_{\bigcup_{n \leq N} A_n} \in K,$$

because the sum of the absolute values of the coefficients is at most one. The sequence (F_N) converges pointwise to $\mathbf{1}_A$. Compactness gives a cluster point in K , and its coordinates force that cluster point to be $\mathbf{1}_A$. Hence $\mathbf{1}_A \in B_\alpha(X)$, so $A \in \Pi_{\alpha+1}^0(X)$. Taking complements yields

$$\Sigma_{\alpha+1}^0(X) = \Pi_{\alpha+1}^0(X),$$

which proves (2) $\Rightarrow$ (4).

Finally, (4) collapses the next additive class:

$$\Sigma_{\alpha+2}^0(X) = \left(\Pi_{\alpha+1}^0(X) \right)_\sigma = \left(\Sigma_{\alpha+1}^0(X) \right)_\sigma = \Sigma_{\alpha+1}^0(X).$$

The Lebesgue–Hausdorff theorem therefore gives $B_{\alpha+1}(X) \subseteq B_\alpha(X)$; the reverse inclusion is automatic. Thus (4) $\Rightarrow$ (3), completing the cycle. $\square$

For $\alpha = 1$, part (B) recovers the source’s σ -space criterion in the metrizable setting. For higher α , it identifies the exact Borel level whose collapse is measured by local and sequential completeness.

6 The full Baire union

Theorem 5 (The $B(X)$ program). *Let X be Tychonoff.*

1. $B(X)$ is always sequentially complete and locally complete.
2. Each strong property in Corollary 2 is equivalent to $B(X) = \mathbb{R}^X$.
3. The equality $B(X) = \mathbb{R}^X$ holds if and only if $\mathbf{1}_A \in B(X)$ for every $A \subseteq X$.

If X is metrizable, the last condition is equivalent to every subset of X being Borel.

Proof. The source proves sequential completeness of $B(X)$ [1, Proposition 3.1]; the general implication from sequential to local completeness gives the rest of (1). Part (2) is Corollary 2.

For the nontrivial direction of (3), suppose every characteristic function belongs to $B(X)$. Every finite-range real function then belongs to $B(X)$. The truncation–quantization sequence (s_n) from the proof of Theorem 1 consists of such functions and converges pointwise to an arbitrary $f : X \rightarrow \mathbb{R}$. Sequential completeness puts f in $B(X)$. The converse is immediate. On a metrizable space, the Lebesgue–Hausdorff theorem identifies characteristic Baire functions with Borel sets of some countable rank. $\square$

7 Upgrade audit, limitations, and novelty check

Eight focused routes were audited.

1. Finite interpolation in the bounded unit box produced the abstract quasi-completion theorem.
2. Successor-rank closure produced the exact sequential criterion.
3. A direct rescaling of telescoping differences failed: pointwise convergence supplies no common rate over an arbitrary X .
4. Disjoint Borel-rank decomposition repaired that failure for metrizable spaces and proved local $\Rightarrow$ sequential completeness.
5. Lebesgue–Hausdorff converted both functional conclusions into exact Borel-hierarchy criteria.
6. Sequential completeness of $B(X)$ completed the full-union program.
7. The same disjointification works on broader domains whenever the appropriate generalized Lebesgue–Hausdorff identification is known; this was not promoted as an unconditional arbitrary-space theorem.
8. A universal nonmetrizable upgrade was tested through functional additive/multiplicative classes. The missing converse from functional measurability to Baire class on arbitrary domains is a genuine obstruction, so the packet stops at the stated scope.

Thus the remaining part of Problem 3.10 is an intrinsic characterization of local completeness of a fixed $B_\alpha(X)$ for arbitrary nonmetrizable Tychonoff X . The packet also does not address the source’s other open problems about Q_f -spaces or special sets of reals.

A bounded novelty search was performed on 11 August 2026. It checked the source arXiv page and published-title result; exact searches for combinations of “ $B_\alpha(X)$ ”, “quasi-complete”, “locally complete”, “ $B_\alpha(X) = B_{\alpha+1}(X)$ ”, and higher Baire-class pointwise spaces; and arXiv searches for

the abstract pointwise quasi-completion statement. The searches returned the source paper, its B_1 results, and general papers on Baire classification, but no statement of Theorems 1 or 4. This supports only bounded novelty confidence.

The most important human checks are: (i) completeness of the bounded box in Theorem 1; (ii) the barrelled weak-space transfer in Corollary 2; and (iii) the transfinite Borel indexing in Theorem 4.

References

- [1] S. Gabrielyan, A. V. Osipov, and E. Reznichenko, *Completeness and reflexivity type properties of $B_1(X)$* , arXiv:2601.00733v2 (2026); Topology Appl., article 109772 (2026).
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